

DATA MODEL

- VISUALIZATION LAYER

INTERACTION LOGIC

- DAX EXPRESSIONS

- MEASURE DEFINITIONS

- **PARAMETER CONTROLS**

Power BI: Advanced Reporting

Architecting Dynamic, App-Like Data Experiences

VISUALIZATION LAYER

INTERACTION LOGIC

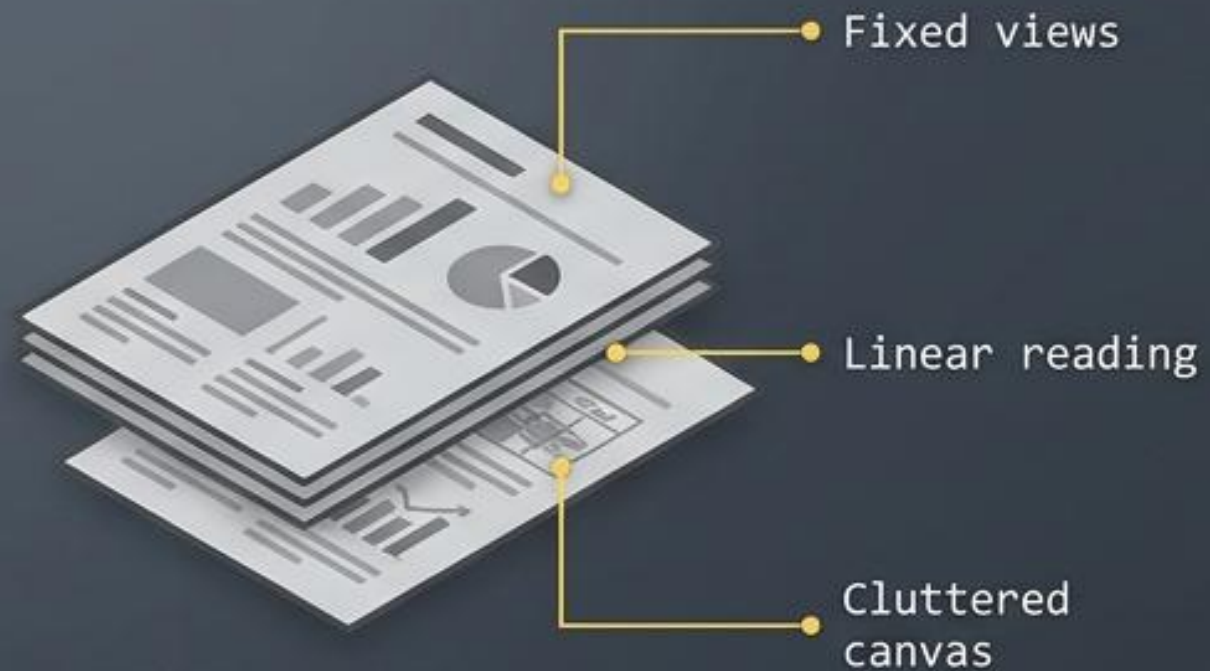
MEASURE DEFINITIONS

PARAMETER CONTROLS

dr Zbigniew Galar

The Reporting Architect's Paradigm

True advanced reporting moves beyond simply displaying data. It requires designing an application-like experience where structure, interactivity, and state management are deliberately controlled.



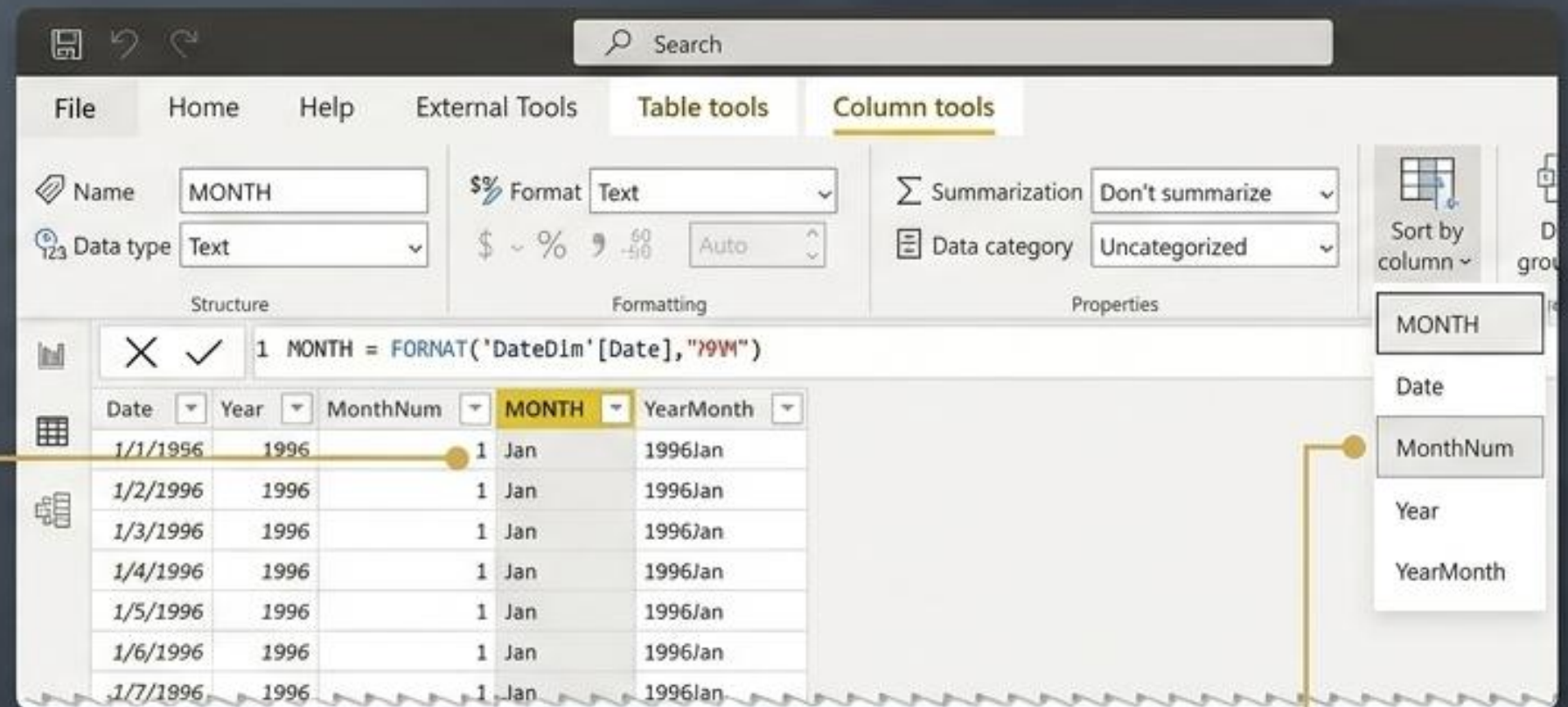
Structural Control: Custom Sorting Rules

Native sorting behaves alphabetically, which breaks chronological data (e.g., Month names).

1. Create a calculated column establishing the desired numerical order (e.g., MonthNum).

2. Select the target text column.

3. Apply the Sort by column override to map the text to the hidden numerical logic.



The screenshot shows the Power BI interface with the 'Table tools' and 'Column tools' tabs. The 'Column tools' tab is active, showing the 'MONTH' column properties. The 'Data type' is set to 'Text'. The 'Format' is set to 'Text'. The 'Summarization' is set to 'Don't summarize'. The 'Data category' is set to 'Uncategorized'. The 'Sort by column' dropdown is open, showing 'MONTH', 'Date', 'MonthNum', 'Year', and 'YearMonth'. 'MonthNum' is selected. The table below shows the data for the 'MONTH' column.

Date	Year	MonthNum	MONTH	YearMonth
1/1/1996	1996	1	Jan	1996Jan
1/2/1996	1996	1	Jan	1996Jan
1/3/1996	1996	1	Jan	1996Jan
1/4/1996	1996	1	Jan	1996Jan
1/5/1996	1996	1	Jan	1996Jan
1/6/1996	1996	1	Jan	1996Jan
1/7/1996	1996	1	Jan	1996Jan

Hierarchical Navigation Diagnostics

These controls only activate when multiple fields exist in a hierarchical arrangement within a field well.

Row 1: The Trident

Expand all down one level in the hierarchy

The "Trident" shows the next level of the fields, while keeping the level(s) above visible.

Category	Net Price
Audio	\$72,420.92
Bluetooth Headphones	\$29,794.40
MP4&MP3	\$13,255.42
Recording Pen	\$29,371.10
Cameras and camcorders	\$236,303.72
Cell phones	\$740,308.51
Computers	\$1,486,938.85
Games and Toys	\$28,301.20
Home Appliances	\$611,439.28
Total	\$3,928,585.63

Expand All Down

Shows the next level of the hierarchy while keeping the parent level visible.

Row 2: Double Arrows

Go to Next Level

Subcategory	Net Price
Air Conditioners	\$38,041.77
Bluetooth Headphones	\$94.40
Boxed Games	\$40.78
Camcorders	\$100,084.35
Cameras & Camcorders Accessories	\$12,712.07
Car Video	\$83,225.67
Cell phones Accessories	\$16,412.26
Coffee Machines	\$57,954.07
Computers Accessories	\$22,244.70
Total	\$3,928,585.63

Go to Next Level

Displays the next layer entirely by itself, stripping away the parent context.

Row 3: Up Arrow



Drill Up

Navigates backward up the structural tree.

The Reactive Canvas: Signal Transmissions

Power BI UI Design Studio: Interaction Specifications

Concept

Visuals communicate by default. To control this network, navigate to Format > Edit Interactions.

Rule 1 (Directionality):
Interactions are configured by first selecting the Source visual, then defining how it impacts the Targets.

Rule 2 (Asymmetry):
Relationships do not have to be reciprocal. Visual A can filter Visual B, while Visual B only highlights Visual A.

Rule 3 (Performance):
Excessive active interactions degrade page rendering speed. Disable unnecessary signals.

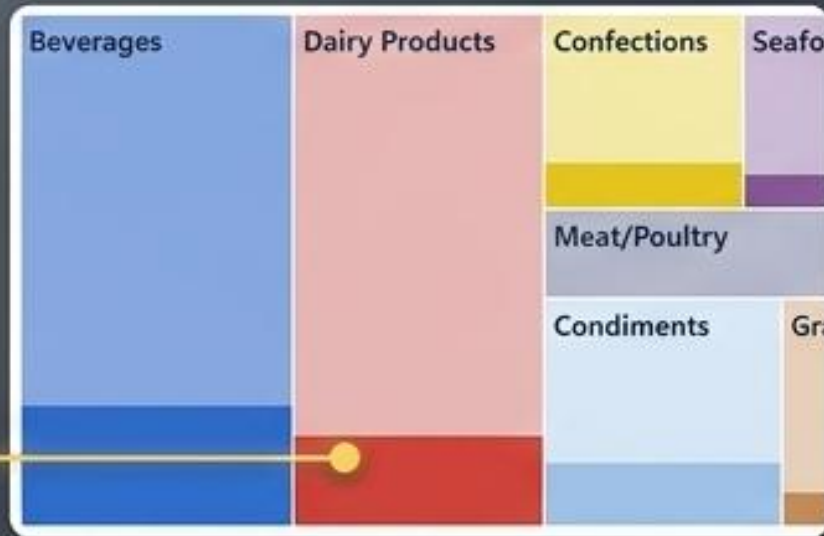
Source Visual: "All Sales by Country"

Interaction Controls Active



Interaction Modes Matrix

Highlight (Default)

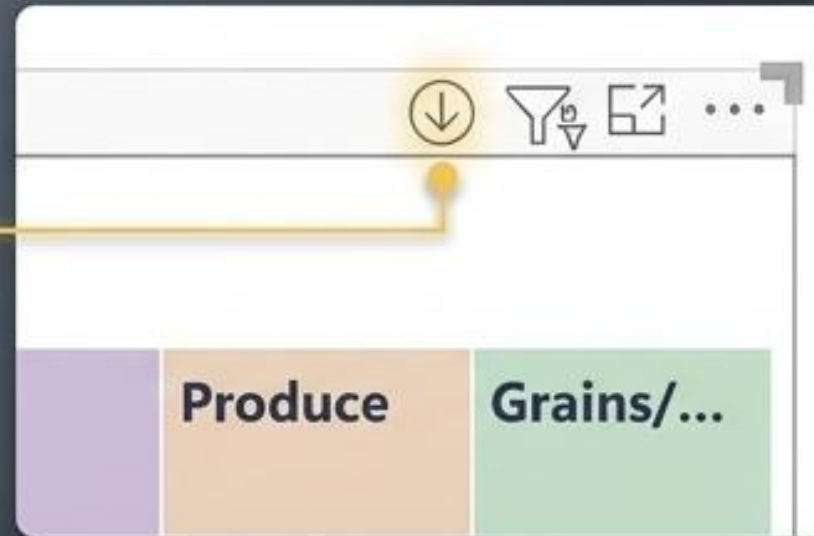


Dims non-relevant data.
Maps do not support
highlighting.



UX Warning: Light colors may become illegible to screen readers when grayed back.

Filter



Removes non-relevant data
entirely, recalculating
the visual's scale to show
only the selected segment.

None



Severs the connection.
The target visual remains
completely static
regardless of source
input.

Canvas Depth: The Selection Pane

Power BI UI Design Studio: Interface & Interaction Specifications

Feature 1 (Layer Order):

Functions identically to presentation software. Controls which elements sit on top of others.

Selection

Layer order	^	v
Button	👁	👁
Button	👁	👁

Feature 2 (Visibility):

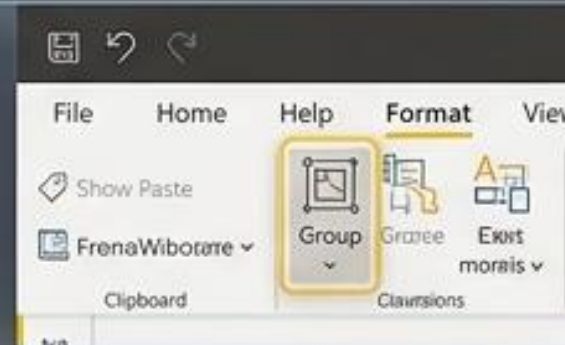
The 'Eye' icon toggles element rendering on/off. Critical for building dynamic menus.

Selection

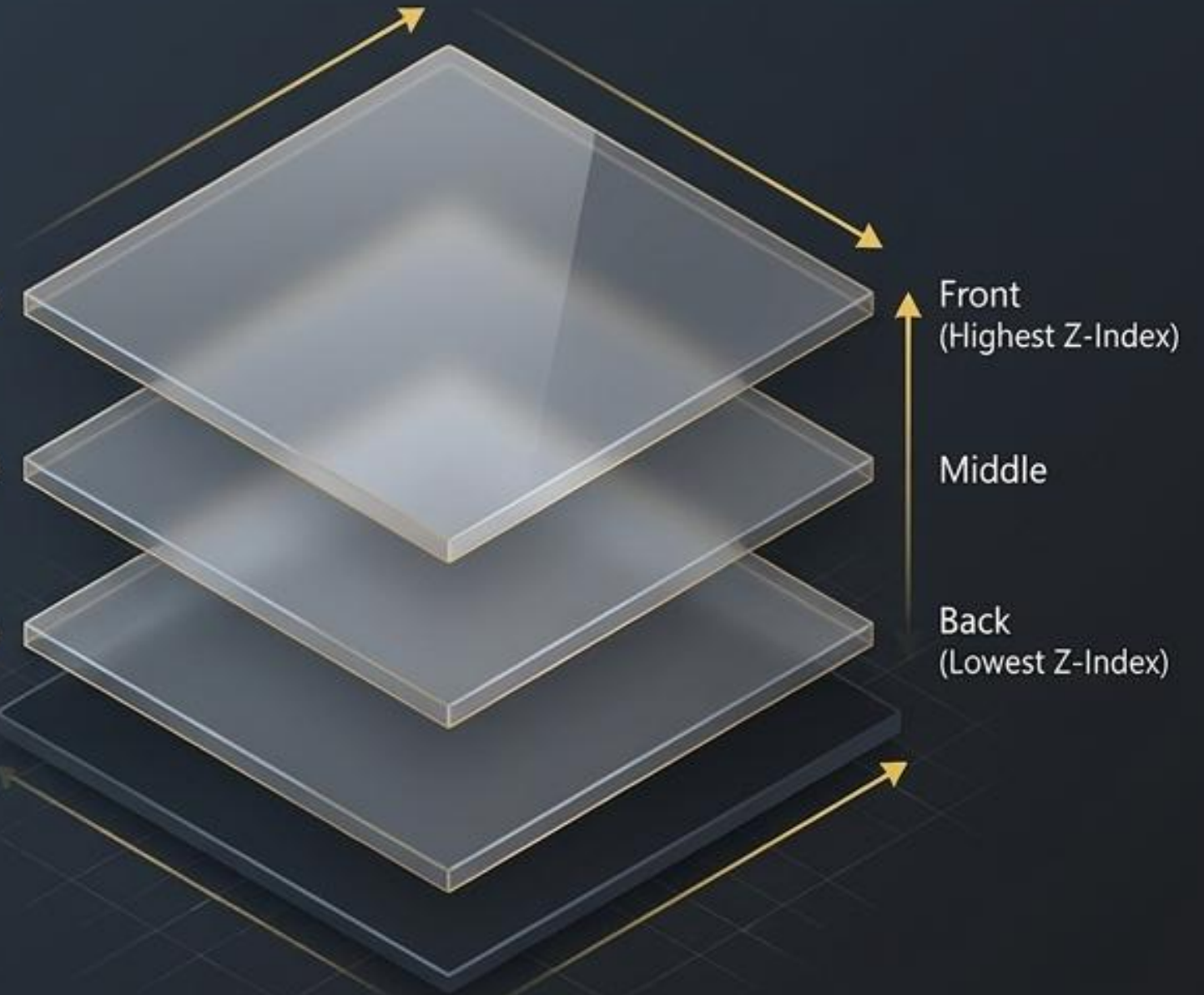
Button	👁
Button	👁

Feature 3 (Grouping):

Format > Group binds elements into blocks for single-click visibility toggles.



Z-Axis Depth



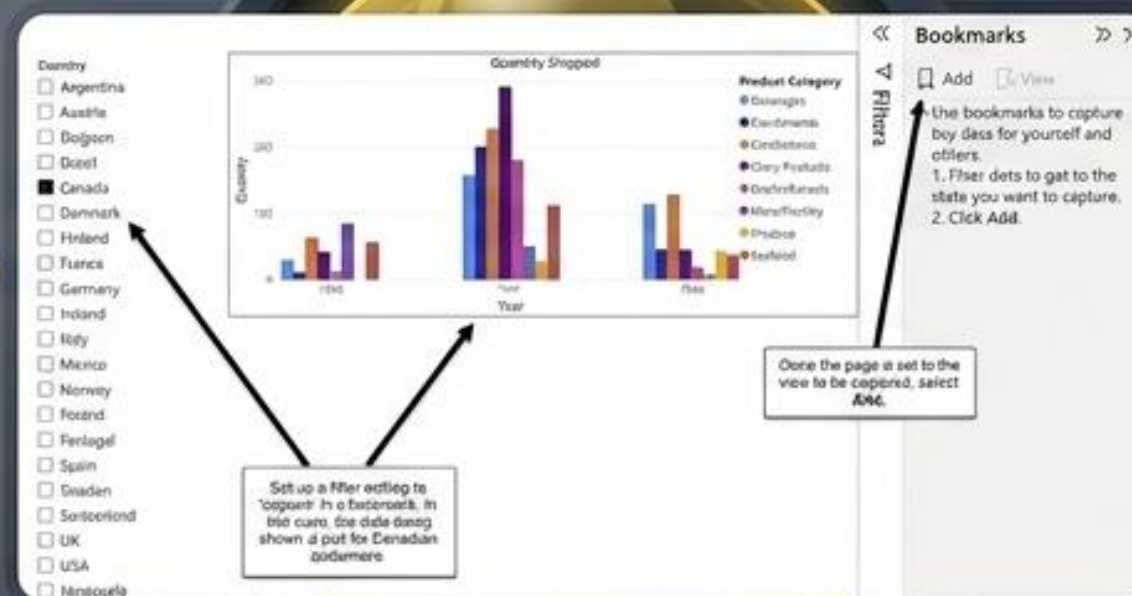
Developer Warning: Renaming a visualization in the Selection pane permanently alters the visual's public title. Renaming buttons or shapes does not.

State Capture: The Bookmark Engine

A bookmark does not just save a page; it freezes the precise configuration of filters, slicers, and visible layers at a specific moment.

The Workflow

1. Configure slicers/filters.
2. Set element visibility.
3. Click 'Add Bookmark'.
4. Rename for clarity.

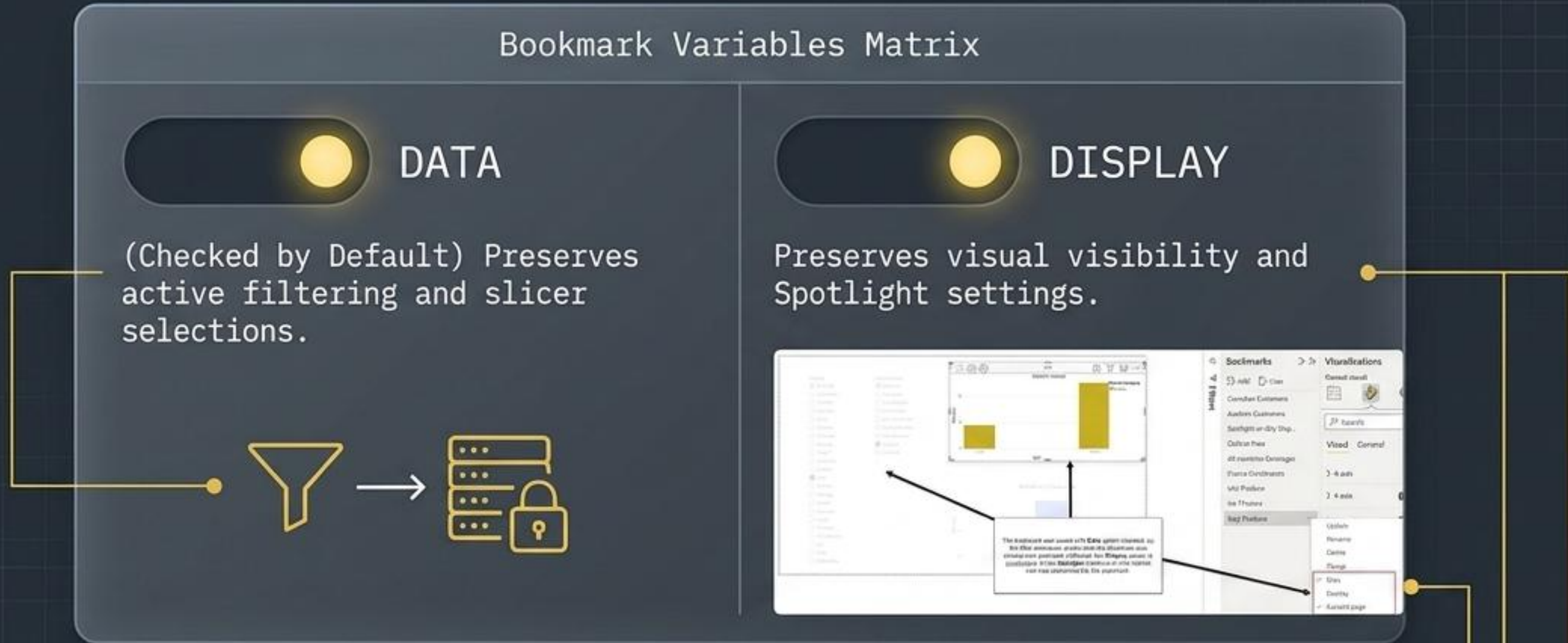


Golden Rule

If the underlying page configuration changes, the bookmark must be manually updated via the ellipsis menu.

Build a Default View first as a safety anchor.

Demystifying Bookmark Variables



The Inheritance Rule

If 'Display' is unchecked when a bookmark is saved/updated, recalling that bookmark will inherit whatever visibility/spotlight settings are currently active on the canvas, rather than forcing its own.

Interactive Triggers: Button Mapping



Implementation

Any button can be assigned an 'Action' type of 'Bookmark'.

The Navigator

For bulk implementations, use the built-in Bookmark Navigator button which automatically binds to existing bookmarks and supports rich formatting.

Dev Environment Note: In Power BI Desktop, buttons require a CTRL + Click to execute. Once published to the Service, a standard click works. Always add descriptive text to guide the user.

Synthesis: Architecting the Tabbed Menu

The Illusion: Building a multi-page app experience without leaving a single Power BI report page.

Step 1 (Z-Axis):

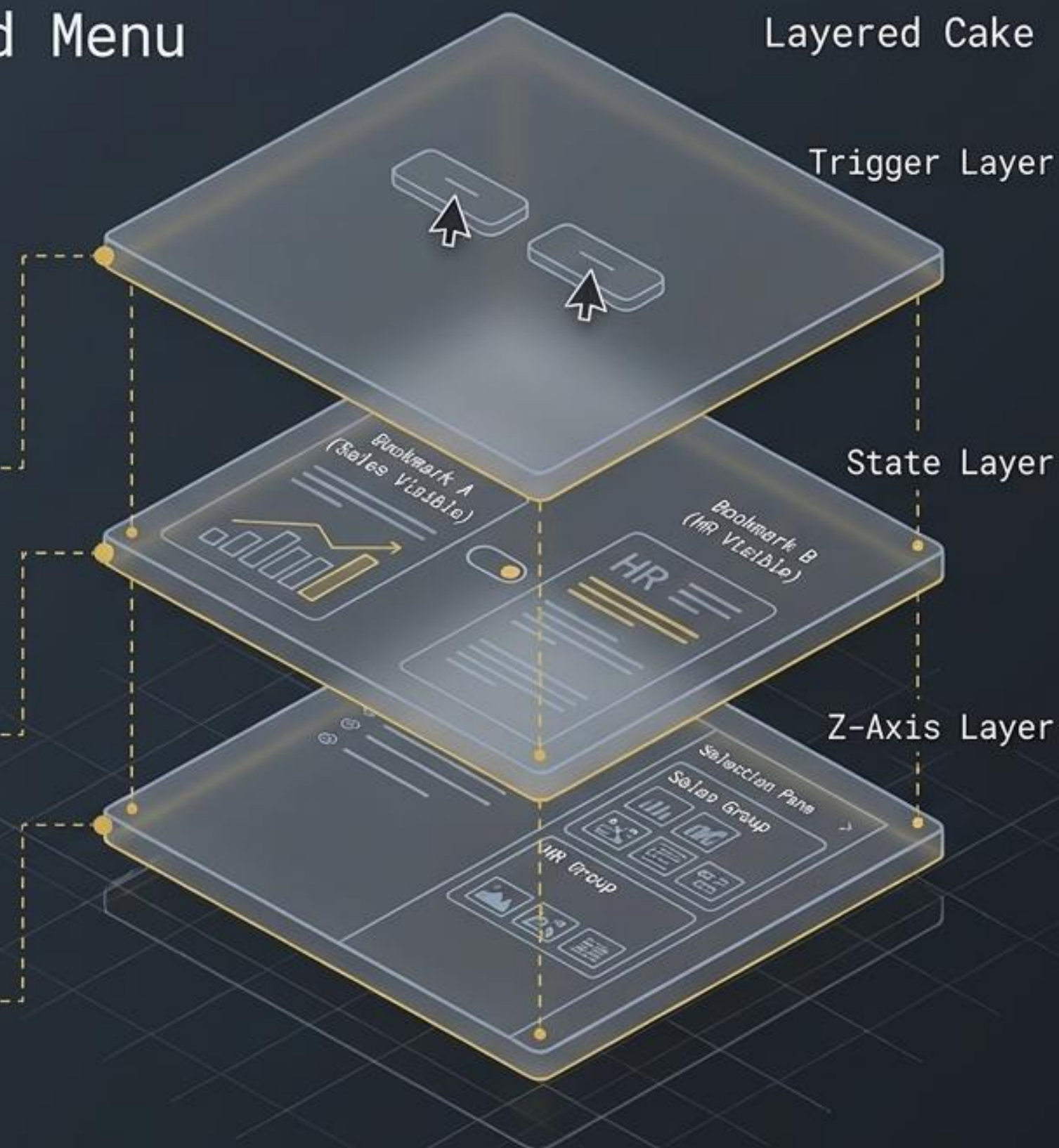
Group distinct sets of visuals in the Selection Pane (e.g., Sales Group, HR Group).

Step 2 (State):

Create Bookmark A with Sales visible / HR hidden. Create Bookmark B with inverted visibility. Ensure Display is checked.

Step 3 (Trigger):

Place two buttons on the canvas. Map Button 1 to Bookmark A, and Button 2 to Bookmark B.



Contextual Intelligence: Micro-Metrics

Enhancing the Default:

Drag additional measures into the Tooltips field well to surface secondary data upon user hover.

X-axis

Add data fields here

Tooltips

NumofProductIDs ✓ ✕

UI Polish: Rename clunky database fields (e.g., NumofProductID) directly for the visual to ensure readability. Adjust background transparency and text color under Format > General.

Architect's Warning: Every measure added to a tooltip requires calculation time. Overloading the tooltip well degrades overall page load performance, even before the user hovers.



The Inception Framework: Tooltip Pages

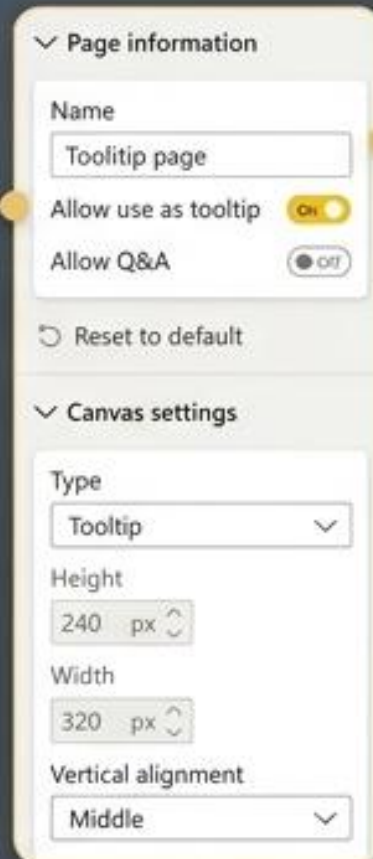
Concept: Embedding a dedicated, hidden report page directly inside a hover state.

1. Create and Hide a new page.

2. Set "Allow use as tooltip" to ON.

3. Set Canvas to **Custom** (e.g., 320x100px).

4. Bind the target visual's tooltip settings to 'Report Page'.



The settings panel is divided into two sections: 'Page information' and 'Canvas settings'. In the 'Page information' section, the 'Name' field is set to 'Tooltip page', 'Allow use as tooltip' is turned ON, and 'Allow Q&A' is turned OFF. A 'Reset to default' button is also present. In the 'Canvas settings' section, the 'Type' is set to 'Tooltip', 'Height' is 240 px, 'Width' is 320 px, and 'Vertical alignment' is set to 'Middle'.

Section	Property	Value
Page information	Name	Tooltip page
	Allow use as tooltip	ON
	Allow Q&A	OFF
	Reset to default	Button
Canvas settings	Type	Tooltip
	Height	240 px
	Width	320 px
	Vertical alignment	Middle



Performance Advantage

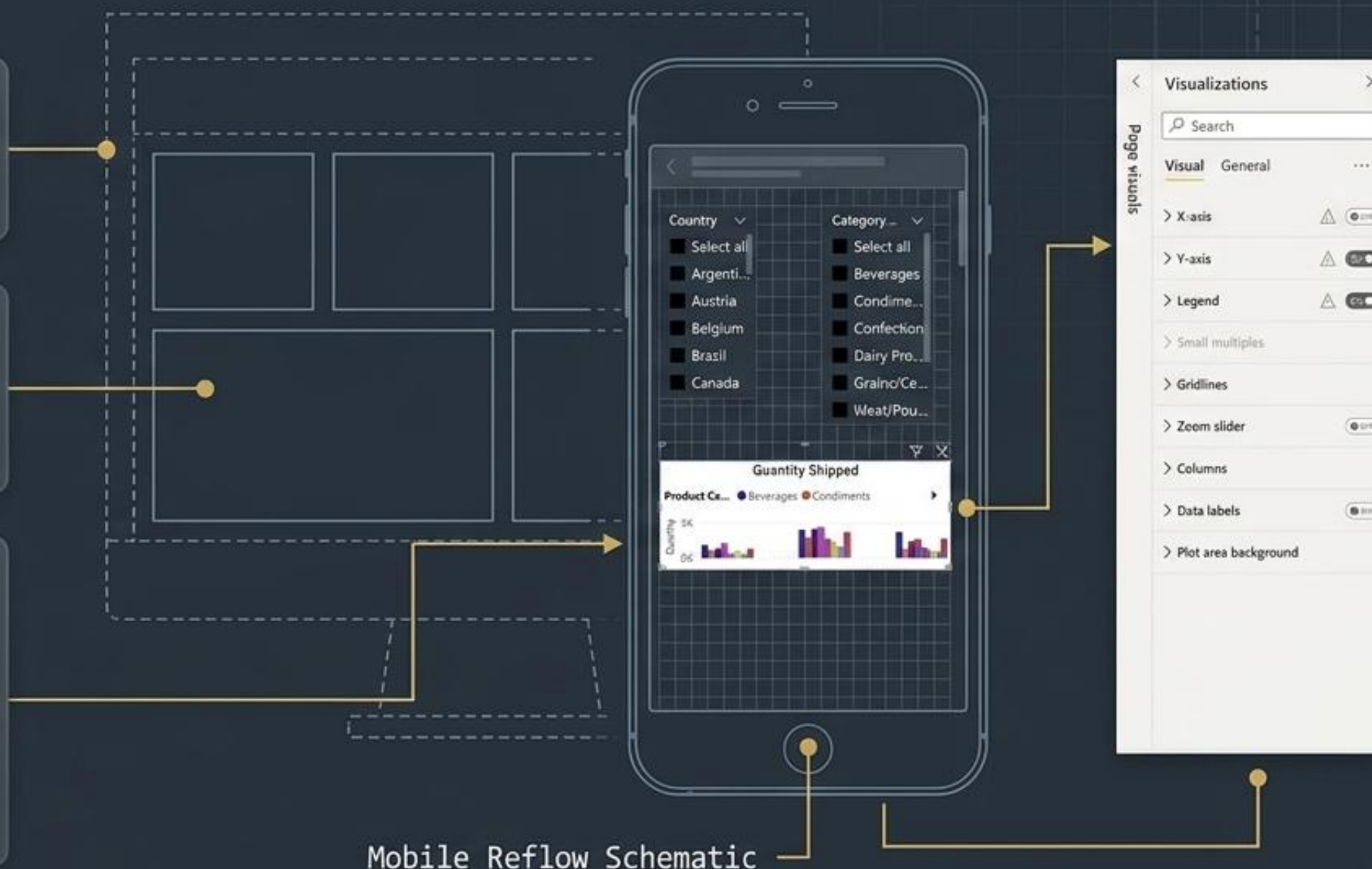
Unlike standard tooltips, measures on a Tooltip Page are strictly lazy-loaded—they do not evaluate until the specific user hover event occurs.

Responsive Delivery: Mobile Architecture

The Engine: View > Mobile layout activates a parallel vertical canvas.

Workflow:
Drag existing desktop visuals from the Page Visuals panel onto the mobile grid.

Adaptive Formatting:
The mobile view auto-adjusts to device sizes. Visual formatting (X/Y axes, legends, gridlines) can be adjusted independently for the mobile view without breaking the desktop layout.



The Architect's Mandate

Structure with Intent

Custom sorts and precise drill-down control protect the data's integrity.

Manage the State

Mastering the Z-axis and Bookmark matrices turns static pages into software.

Context on Demand

Using targeted tooltips and interaction signals reveals complexity only when requested.

Exceptional Power BI design hides technical complexity behind an effortless, intuitive user interface.